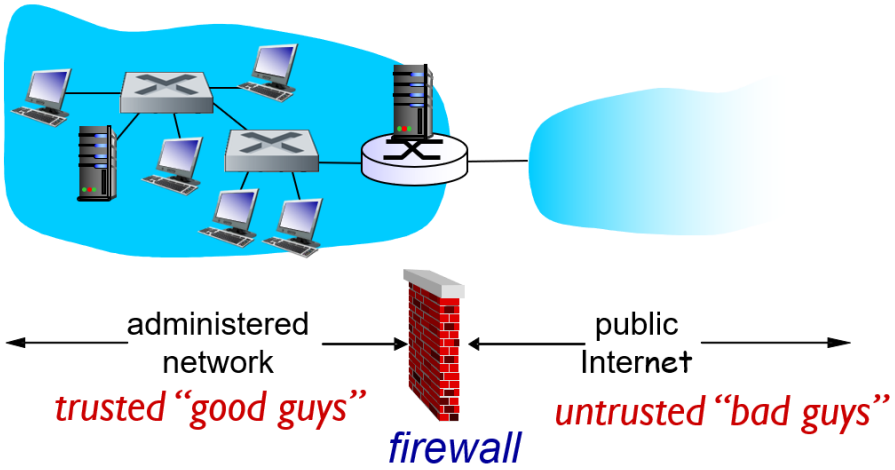
Traditional Network Security Models

Perimeter Security

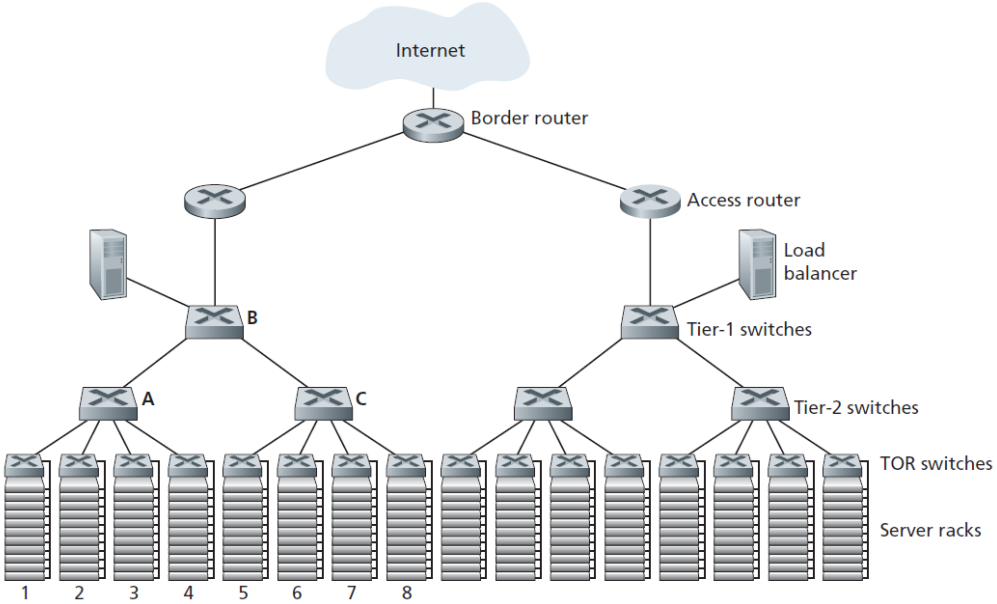
- Perimeter-based network security models date back to early mainframe era
- Large mainframe computers located in physically secure “machine rooms”
- Perimeter-based physical security is still useful but may not be sufficient



Perimeter Security

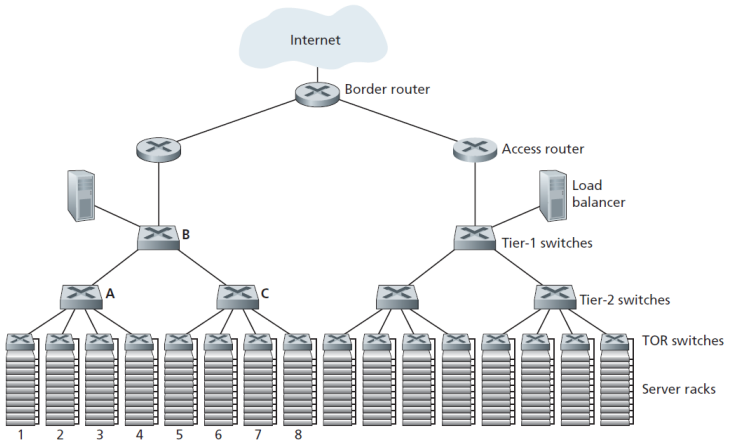
- Today's data centers are the modern equivalent of machine rooms
- Perimeter-based physical security may no longer be sufficient

Microsoft Chicago data center



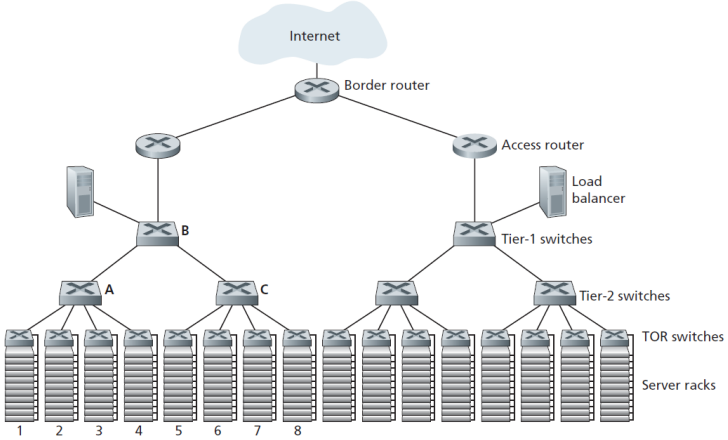
Perimeter Security

- Today's data centers are the modern equivalent of machine rooms
- Perimeter-based physical security may no longer be sufficient
  - Deployed at a handful of well-defined ingress and egress points
  - It relies on the assumption that everything on internal network is trusted



Perimeter Security

- Remote employees, mobile users, cloud computing solutions blur the distinction between “internal” and “external”
- Insiders, whether intentionally malicious or just careless, may present a very real security threat
- The potential for sophisticated cyberthreats to penetrate perimeter defenses, in which case they would then have free passage on the internal network

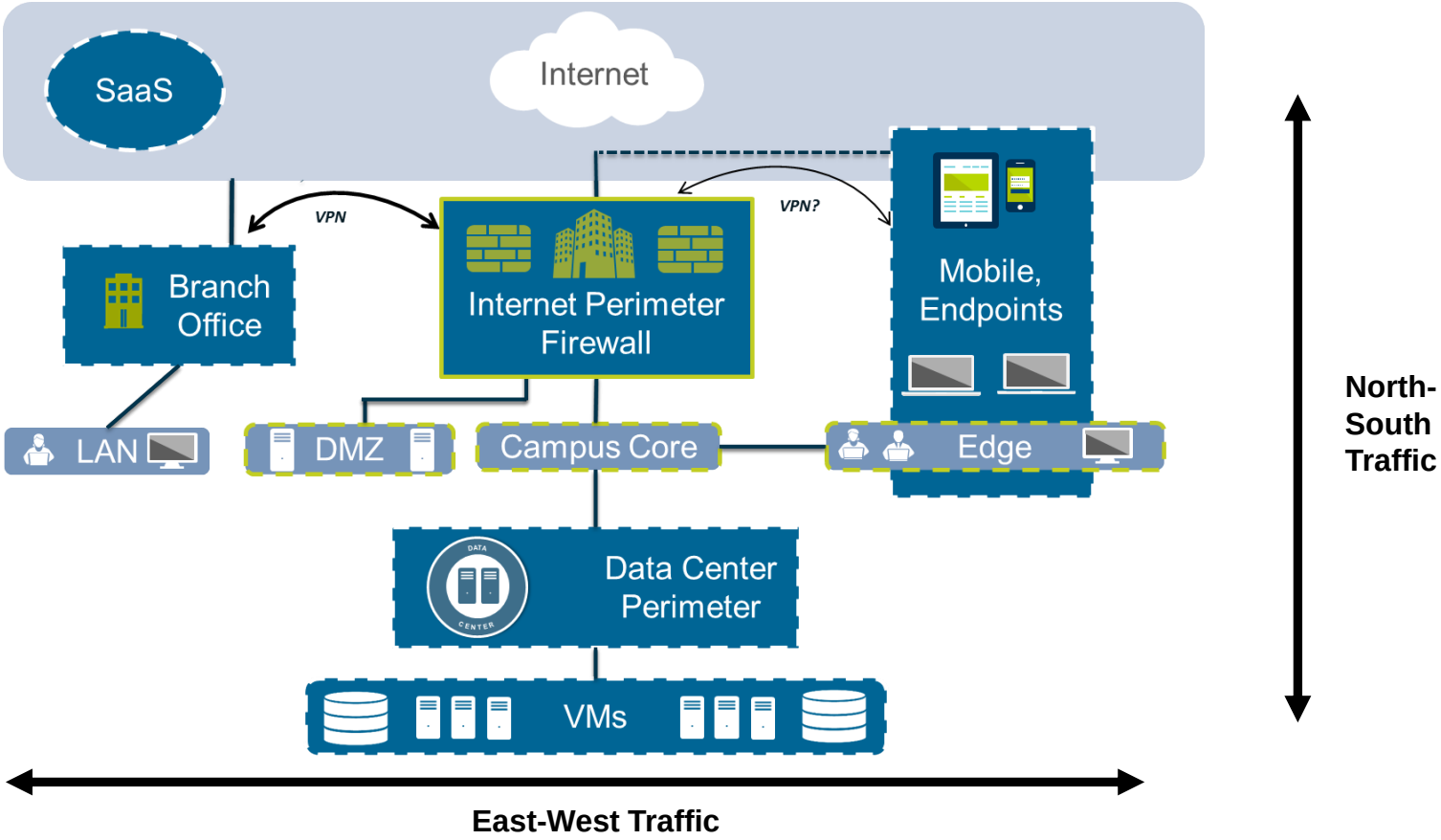


Zero Trust Model

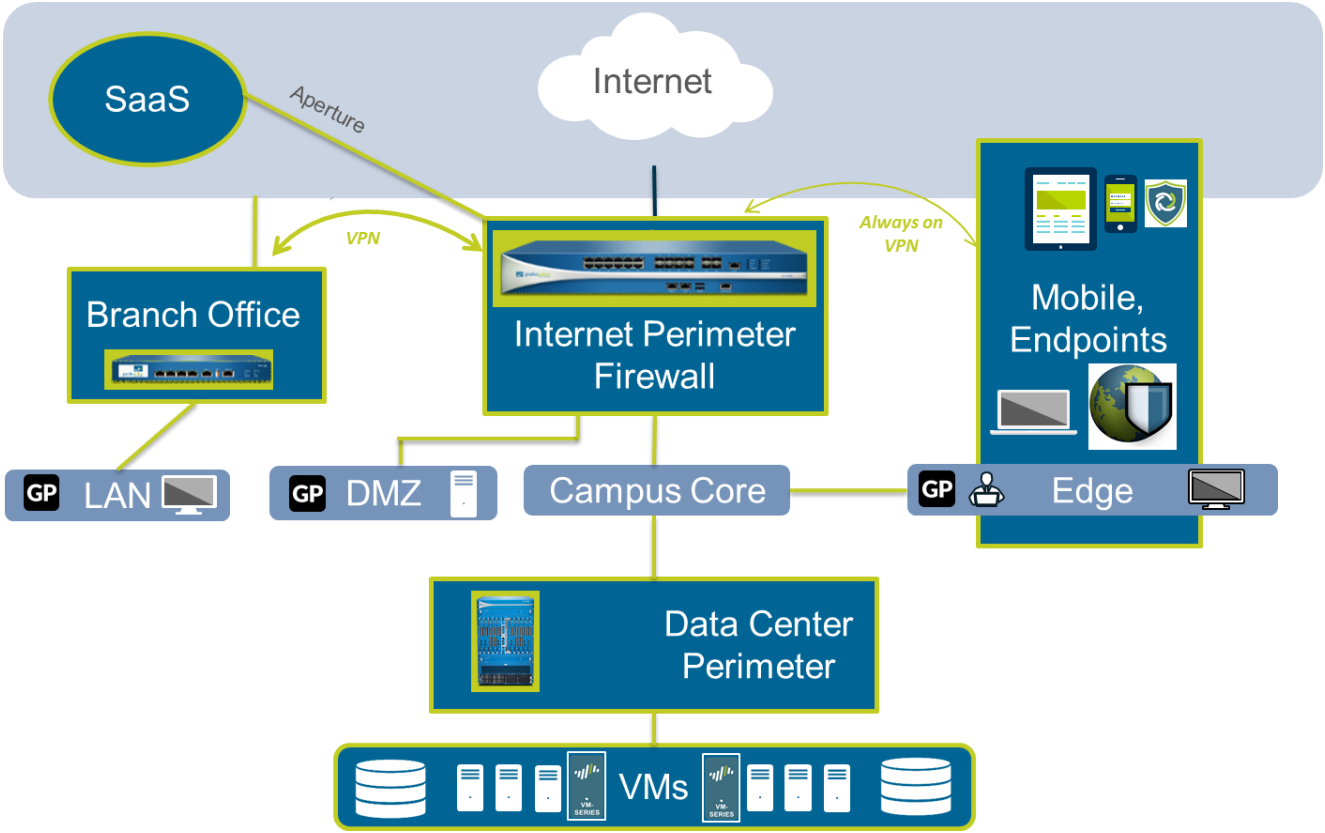
Zero Trust Model

- There is no default trust for any entity
- Move security functionality closer to the pockets of resources that require protection
- Improved ability to secure mobile device networks, BYOD, infrastructure virtualization, and cloud computing

Data Flows in an Open Network



Data Flows in an Open Network



Zero Trust Model

- It is intended to remedy the deficiencies with perimeter-centric strategies and the legacy devices and technologies
- Three important principles:
  - Ensure that resources are accessed securely, regardless of location
  - Adopt a least privilege strategy and strictly enforce access control
  - Inspect and log traffic
- Implications
  - Establish trust boundaries in the internal computing environment
  - Move security functionality closer to resources that require protection
  - Enforce security policy regardless of the origin and communications traffic

Unified Threat Management?

- Unified Threat Management can be considered as a precursor of Next Generation Firewalls
- Integrated device that combines several security functions
  - Anti-virus
  - Anti-spyware
  - Firewall
  - Content filtering
  - URL filtering
  - Intrusion Protection System
  - DDoS detection…

Integrated Approach to Threat Prevention

	Delivery	Exploitation	Installation	C2	Act on Objective
App-ID	Block high risk applications			Block C2 on non-standard ports	Prevent exfiltration and lateral movement
URL Filtering	Block known malware sites			Block malware, fast-flux domains	
Vulnerability		Block the exploit			Prevent lateral movement
Anti-spyware				Block spyware, C2 traffic	
AV			Block malware		Prevent lateral movement
Traps	Monitor allowed processes and executables	Prevent the exploit	Prevent malicious .exe from running		
File Blocking			Prevent drive-by downloads		Prevent exfiltration and lateral movement
DoS/Zone		Prevent evasions			Prevent DoS attacks
WildFire	Identify malware		Detect unknown malware	Detect new C2 traffic	

Next-Generation Security Platform and Architecture

Topics

- Security platform overview
- Next-generation firewall architecture
- Zero Trust security model
- Public cloud security
- Firewall offerings

Objectives

- Explain the motivation for Next-Generation Firewalls (NGFWs)
- Describe advantages of NGFWs
- Describe data and control planes of modern security appliances
- Become familiar with a NGFW architecture

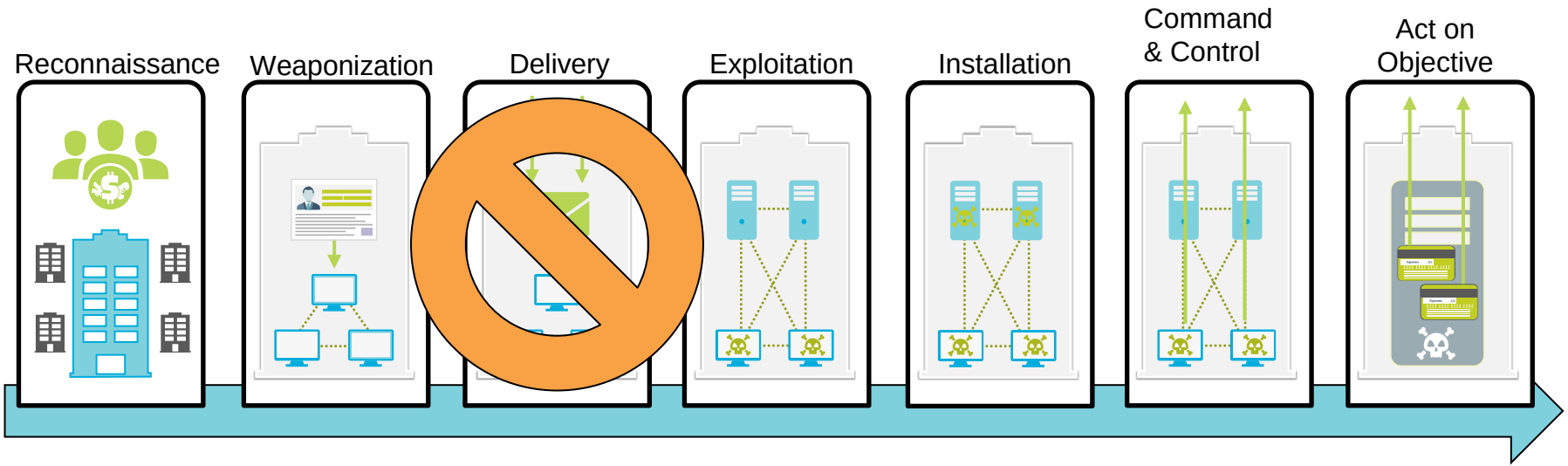
Security Platform Overview

Next-Generation Firewall (NGFW)

- Traditional firewalls make decision based on (only) IP addresses and port
- “A more *intelligent* firewall is an application-aware firewall or *next-generation firewall*.
  - They operate at a higher level
  - Identify applications that send packets through the firewall
  - Make decisions about the application based on granular rule settings
  - Identify applications (e.g., Skype, Facebook chat, etc.) by using predefined signatures, header inspection, or payload analysis
  - Can learn new applications by watching how they behave and creating a baseline of normal behaviors so that an alert can be raised if the application deviates from baseline

Page 282, "Security+ Guide to Network Security Fundamentals."

Cyber Attack Lifecycle



Stop the attack at any point!

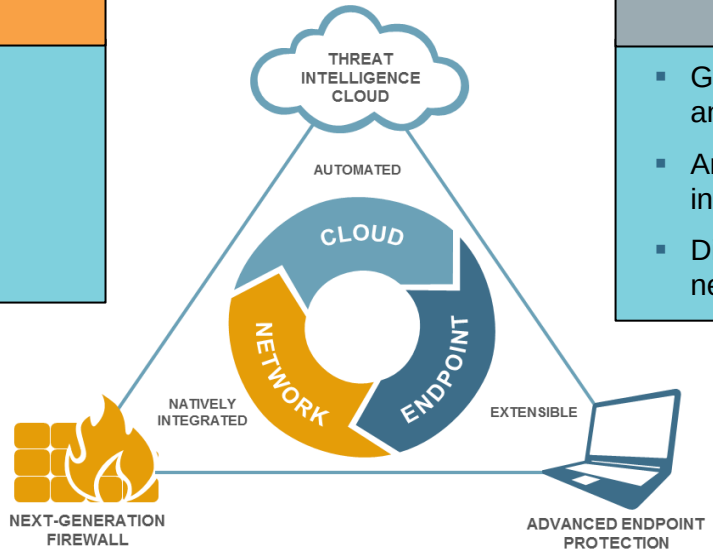
Next-Generation Security Platform

Next-Generation Firewall

- Identifies and inspects all traffic
- Blocks known threats
- Sends unknown to cloud
- Extensible to mobile and virtual networks

Threat Intelligence Cloud

- Gathers potential threats from network and endpoints
- Analyzes and correlates threat intelligence
- Disseminates threat intelligence to network and endpoints



Advanced Endpoint Protection

- Inspects all processes and files
- Prevents both known and unknown exploits
- Integrates with cloud to prevent known and unknown malware

Next-Generation Firewall Architecture

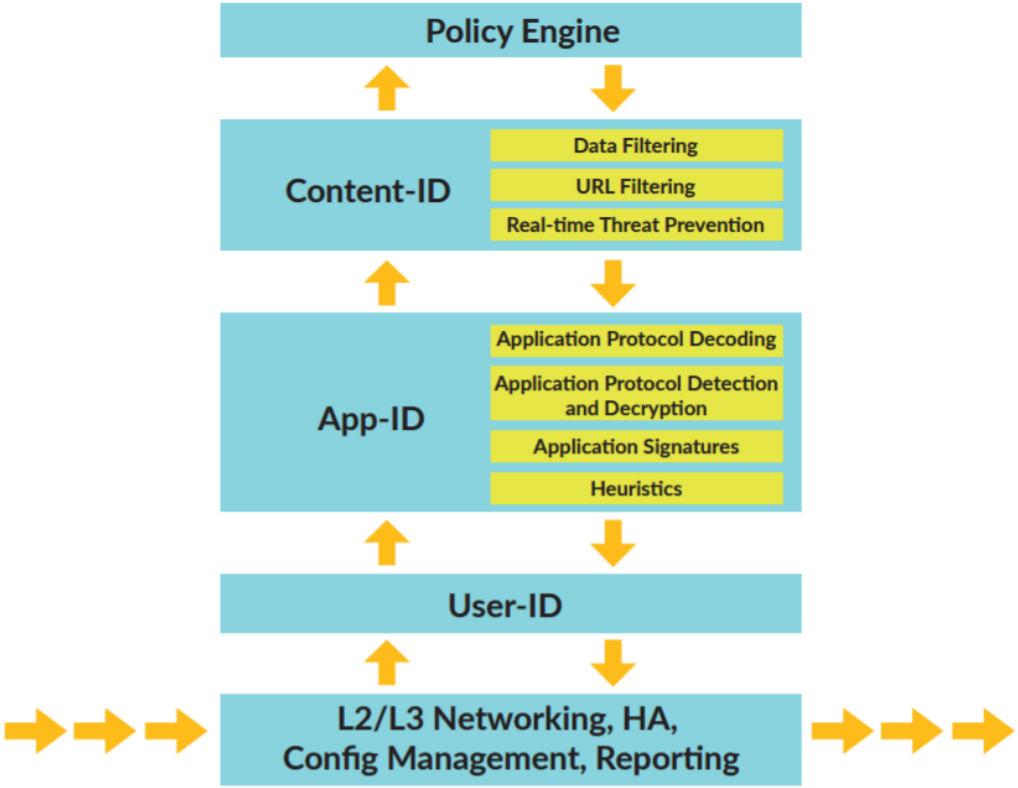
Use Case: Palo Alto Networks Architecture

Single pass:

- Operations per packet:
  - Traffic classification with App-ID technology
  - User/group mapping
  - Content scanning – threats, URLs, confidential data
- One single policy (per type)

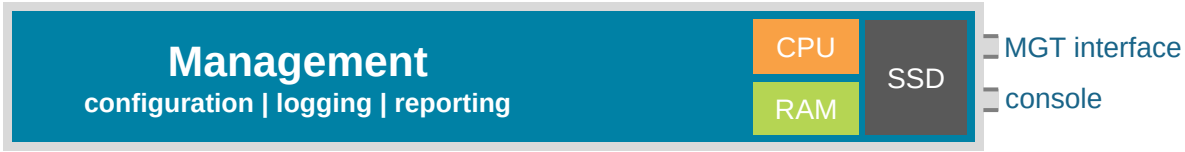
Parallel processing:

- Function-specific parallel processing hardware engines
- Separate data/control planes

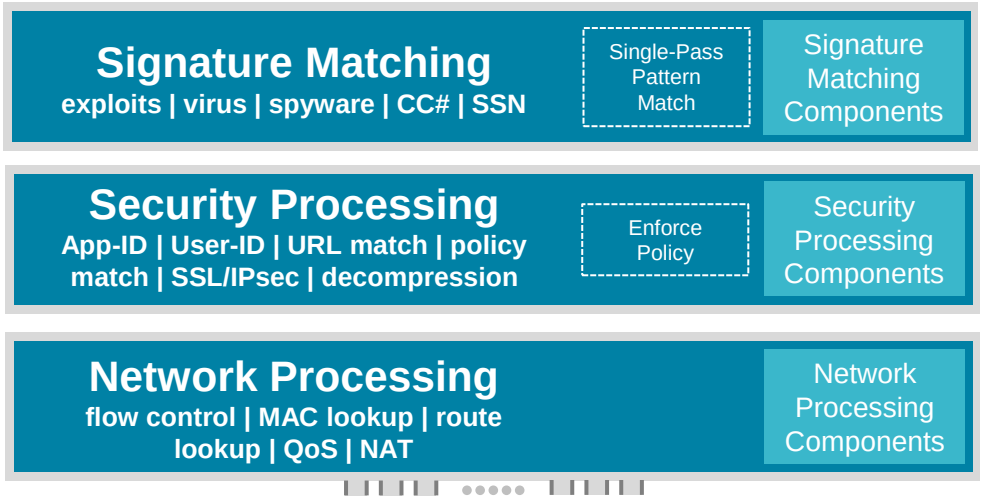


Use Case: Palo Alto Networks Architecture

Control Plane



Dataplane



Hardware component types/sizes per layer vary per firewall model.

Control Plane | Management
Provides configuration, logging, and report functions on a separate processor, RAM, and hard drive

Signature Matching
Stream-based, uniform signature match including vulnerability exploits (IPS), virus, spyware, CC#, and SSN

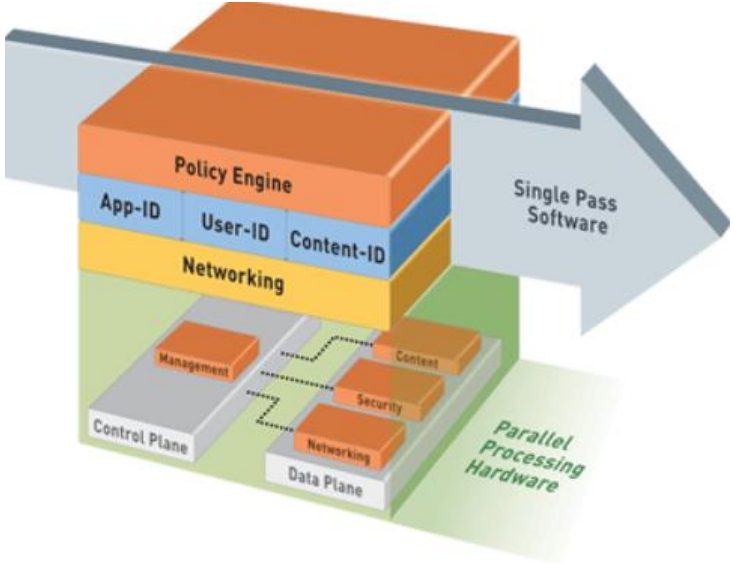
Security Processing
High-density parallel processing for flexible hardware acceleration for standardized complex functions

Network Processing
Front-end network processing, hardware-accelerated per-packet route lookup, MAC lookup, and NAT

Use Case: Palo Alto Networks Architecture

- The **control plane** It is the ‘brain’ of the device. It determines:
  - How packets are routed (to what output interface the packet will be forwarded)
  - What security features must be applied to a packet
  - Provides configuration, logging, and report functions on a separate processor, RAM, and hard drive

- The **data plane** performs the ‘heavy duty’. It performs:
  - Signature matching (e.g., searching for malware)
  - Security processing (e.g., encryption, decryption)
  - Network processing (e.g., NAT)



Use Case: Palo Alto Networks Architecture

- Gartner review (research and advisory IT firm)
- This quadrant analysis focuses on
  - New features that protect customers from emerging threats
  - Expert capability rather than treat the firewall as a commodity
  - Track record of avoiding vulnerabilities in their security products.

Figure 1. Magic Quadrant for Enterprise Network Firewalls

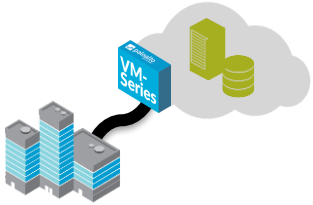
Source: Gartner (October 2018)

https://www.gartner.com/reviews/market/enterprise-network-firewalls/vendor/palo-alto-networks

Public Cloud Security

Public Cloud Security Overview

- Protect your public cloud deployment just as you would your data center.

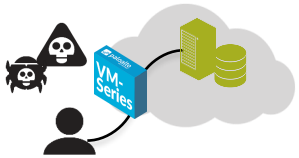


Hybrid

Securely deploy applications in your data center or in the cloud

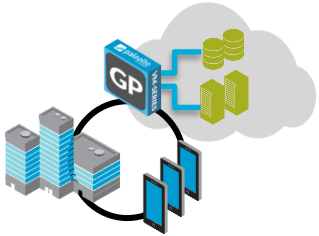
Segmentation

Separate data and applications for compliance and security



Internet Gateway

Protect Internet-facing applications



Remote Access

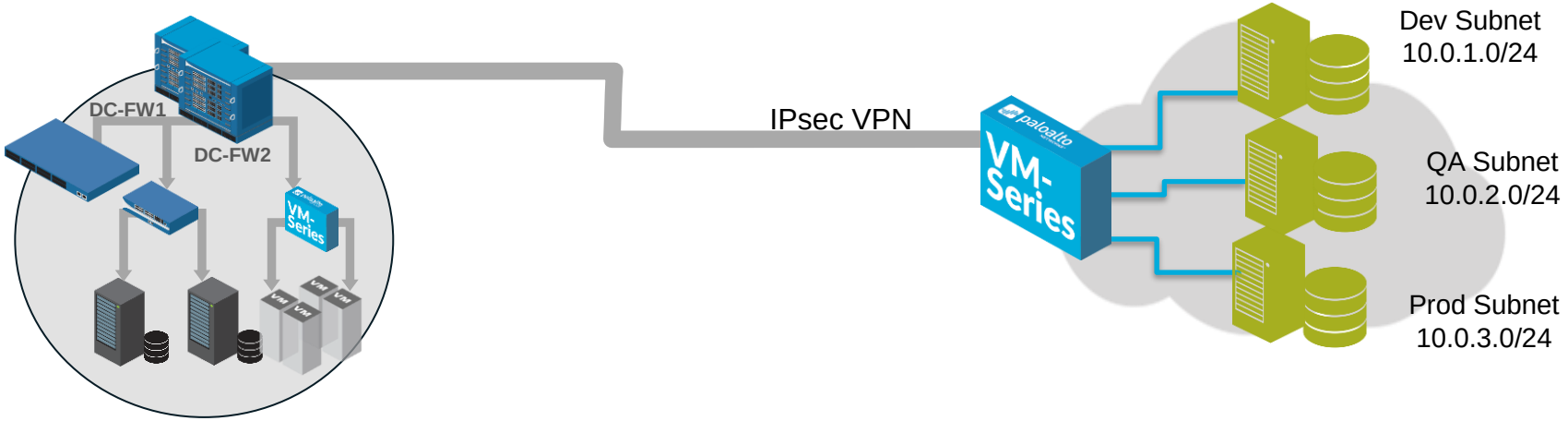
Consistent policy on the network, in the cloud, on devices

NGFWs

- Today's NGFW needs not to be physical devices; instead options include virtual machines (VMs)
- Manage all aspects of the VM-Series – from configuration to policy to reporting – from a centralized location
- Security policies can be consistently enforced across virtualized and physical firewalls

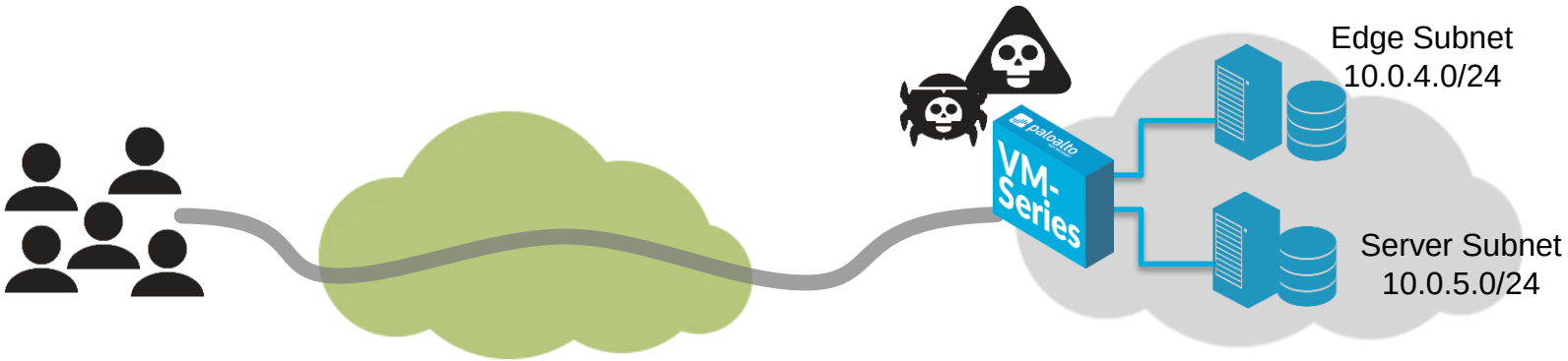
Application Segmentation: Expands upon Hybrid

- Maintain separation between data and applications for security and compliance
- Control which applications can communicate with each other
- Protect traffic within the VPC/vNet and traversing each subnet
- Prevent threats from moving laterally (e.g., avoid malware to move from one subnet of your internal network to another subnet, or from one server to another server)



Internet-Facing Applications: Leverage Perimeter Controls

- Traditional perimeter security strengths apply:
  - Visibility: Classify all traffic based on application identity
  - Control: Enable those applications you want, deny those you don’t
  - Protect: Block known and unknown threats
  - Authorize: Grant access based on user identity



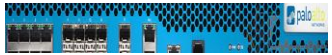
Firewall Equipment

Physical Platforms

Next-Generation Firewalls

PA-5200 Series

PA-5000 Series



PA-3000 Series

PA-7000 Series

PA-800 Series

PA-500

PA-220

PA-200

M-100

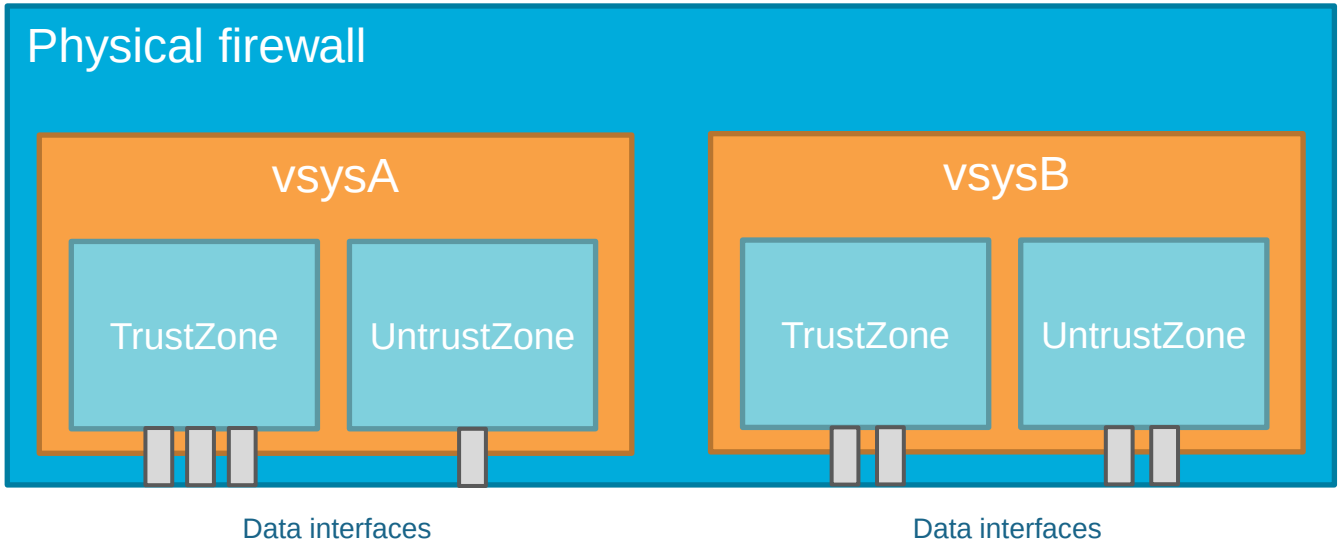
M-500/WF-500

VM-Series Models and Capacities (PAN-OS® 8.0)

Performance and Capacities	VM-700	VM-500	VM-300	VM-100	VM-50
Firewall throughput (App-ID enabled)	**20Gbps**	**10Gbps**	**4Gbps**	**2Gbps**	**200Mbps**
Threat prevention throughput	**10Gbps**	**5Gbps**	**2Gbps**	**1Gbps**	**100Mbps**
Max sessions	10,000,000	2,000,000	800,000	250,000	50,000
Dedicated CPU cores	2, 4, 8, 16	2, 4, 8	2, 4	2	2
Dedicated memory (minimum)	48GB	16GB	9GB	6.5GB	4.5GB
Dedicated disk drive capacity (minimum)	60GB	60GB	60GB	60GB	32GB

Virtual Systems

- Separate, logical firewalls within a single physical firewall
- Creates an administrative boundary
- Use case: multiple customers or departments



Cisco NGFW

ASA 5500-X with FirePOWER Services

- For small to medium business, branch office
- Firewall throughput from 256 to 1750 Mbps
- Threat inspection from 125 to 1250 Mbps
- Stateful firewall, Application Visibility and Control, NGIPS, Advanced Malware Protection, URL filtering

Firepower 2100 Series

- For Internet edge to data center environments
- Firewall throughput from 2.0 to 8.5 Gbps
- Threat inspection from 2.0 to 8.5 Gbps
- Stateful firewall, Application Visibility and Control, NGIPS, Advanced Malware Protection, URL filtering

Firepower 4100 Series

- For Internet edge, high-performance environments
- Firewall throughput from 12 to 30 Gbps
- Threat inspection from 10 to 24 Gbps
- Stateful firewall, Application Visibility and Control, NGIPS, Advanced Malware Protection, URL filtering, DDos

Firepower 9000 Series

- For service provider, data center
- Firewall throughput up to 225 Gbps
- Threat inspection up to 90 Gbps
- Stateful firewall, Application Visibility and Control, NGIPS, Advanced Malware Protection, URL filtering, DDoS

- https://www.cisco.com/c/en/us/products/security/firewalls/index.html#~stickynav=1